



Introduction to Blockchain

CSE-355

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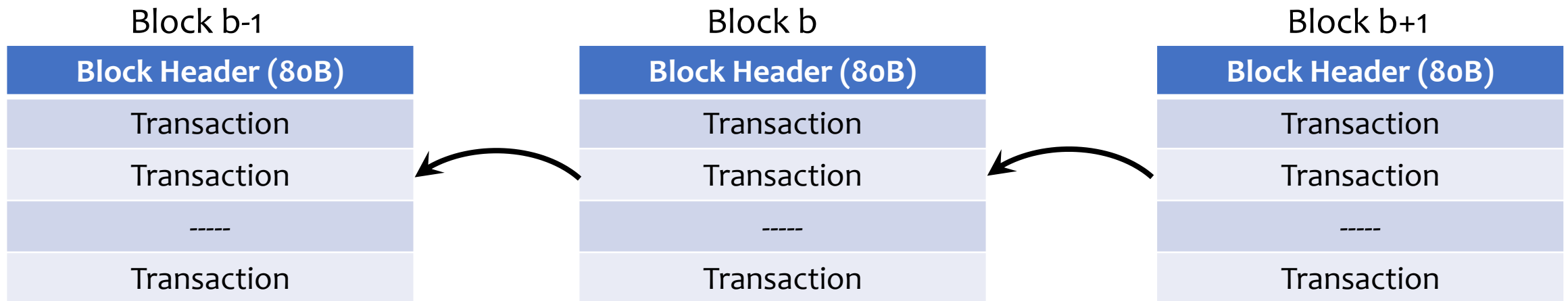
SMCS, IBA Karachi

Spring 2026

Lecture 13

Tuesday, Mar 3, 2026

Blockchain Block Header



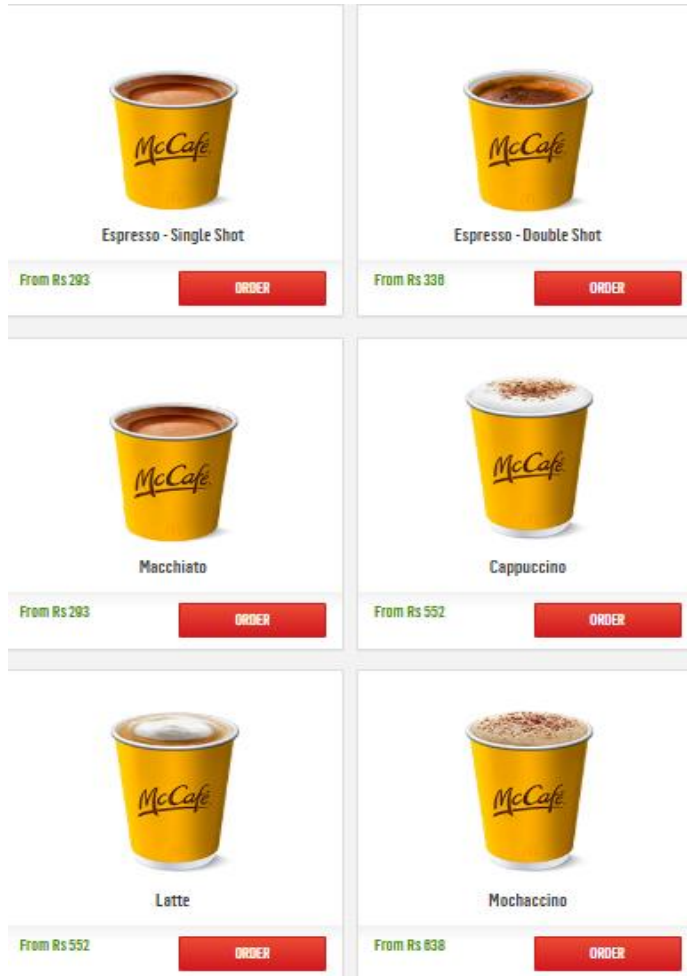
There are 6 fields in the block header:

1. Version (4B) – for exchanging messages and signaling the protocol rules
2. Previous block hash (32B) – for linking current block to its parent block
3. Merkle root hash (32B) – for summarizing & maintaining integrity of transactions in the block
4. Timestamp (4B) – for recording block creation time; up to 2 hour variation is acceptable because Bitcoin nodes are not concerned about timing synchronization
5. Target_Hash (4B) – for defining the difficulty of finding the block header hash; adjusts every 2016 blocks (~ two weeks) to ensure 10 minute average block generation time
6. Nonce (4B) – for iterating values during hash attempts to find a valid block header

$\text{Block_Header} = [\text{Version} + \text{PreviousBlockHash} + \text{MerkelRootHash} + \text{Timestamp} + \text{DifficultyTarget} + \text{Nonce}]$

$\text{Block_Header_Hash} = \text{SHA256}(\text{SHA256}(\text{Block_Header}))$

Can We Pay for Coffee in BTC?



- Bitcoin price are too high for a single cup of coffee
- Average block generation time is 10 minutes, so do we want to wait this long every time for a coffee payment to go through?
- Do we really want every cup of coffee we drink broadcast to the whole world?
- Billions of coffee cups are consumed every day so can bitcoin process all these coffee transactions?

Bitcoin Statistics: Jan 22, 2026

- Bitcoins in circulation: **19.98 Million** (over \$2T market cap)
- Total will be **21 Million** (last will be mined about in the year **2140**)
- Bitcoins left to be *mined*: **1.02 Million** (Total issued: **95.1%**)
- Blockchain Size: **715 GB**
- New Bitcoin Blocks per day (presently): **144** (on average)
- New Bitcoins per day (presently): **450** ($144 \times \underline{\mathbf{3.125}}$)
- About **4 Million** Bitcoins were lost forever (estimate)
- Transactions per second: **7** (VISA has about **24k**)
- Bitcoin Price: **\$ 66k**

Does Bitcoin look like a peer-to-peer electronic cash system?

Bitcoin – Decentralized Digital Currency

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshiin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

Satoshi envisioned Bitcoin as a peer-to-peer digital currency for all kind of online payments



<https://news.bitcoin.com/10-years-ago-bitcoins-genesis-block-changed-the-course-of-history/>

Bitcoin as a Micropayment Platform?

Is there a way to make bitcoin a micropayment platform without sacrificing the decentralization or security?

Bitcoin Scripts Revisited

What Can Bitcoin Opcodes Do?

- **Check signatures**
Opcodes like OP_CHECKSIG verify that a transaction is authorized by the owner of a private key –
- **Combine conditions**
Opcodes like OP_IF, OP_ELSE, and OP_ENDIF allow multiple spending conditions (e.g., *Alice OR Bob can spend*)
- **Require multiple approvals**
Opcodes like OP_CHECKMULTISIG allow *M-of-N* signatures (e.g., 2 out of 3 people must approve)
- **Lock funds by time**
Opcodes like OP_CHECKLOCKTIMEVERIFY (CLTV) can prevent spending until a certain date or block height –
- **Lock funds by delay**
Opcodes like OP_CHECKSEQUENCEVERIFY (CSV) require waiting some time *after* a transaction is created

What is possible with Bitcoin Scripts?

Lock BTC with different spending conditions inside Bitcoin transactions

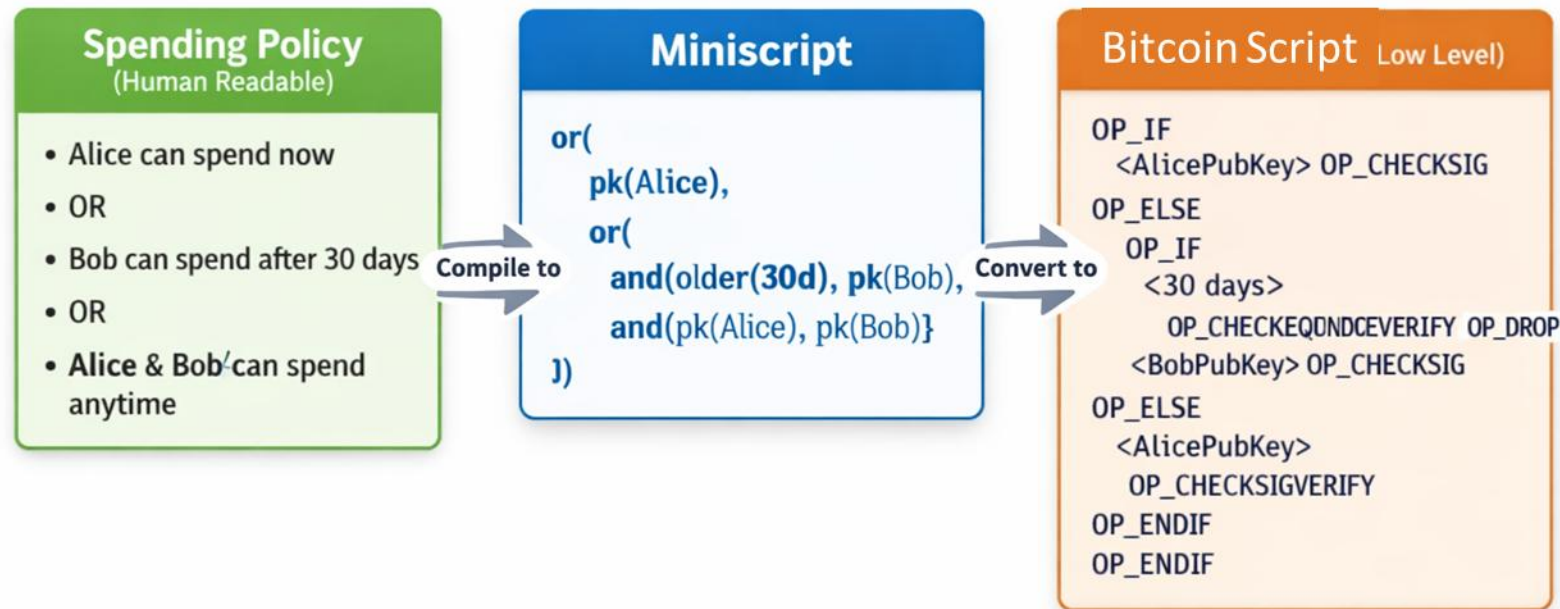


- Time locks
 - Spending via absolute-time-based lock (nLockTime & CheckLockTimeVerify (CLTV))
 - Spending via relative-time-based lock (nSequence & CheckSequenceVerify (CSV))
- Hash locks
 - Spending via Hash locks
 - Spending via Hashed Time lock Contract (HTLC)

But scripts are low-level and hard to write
Bitcoin Script is stack-based and not designed to be human-friendly

Miniscript

- Miniscript is like **TypeScript** for Bitcoin
- Just as TypeScript turns into JavaScript so browsers can read it, Miniscript turns into **Bitcoin Script** so the blockchain can read it
- It allows developers to build "Smart Contracts" on Bitcoin without needing a PhD in Opcodes



Comparison

RAW Bitcoin Script

- Hard to read
- Easy to mess up
- Not intuitive
- Hard to verify

Miniscript

- Structured
- Readable
- Matches the policy
- Easier to reason about

Bitcoin Transactions – UTXO Model

- Bitcoin transactions are created using unspent transaction output (UTXO model)
 - A UTXO has well-defined inputs (one or more) and outputs (one or more)



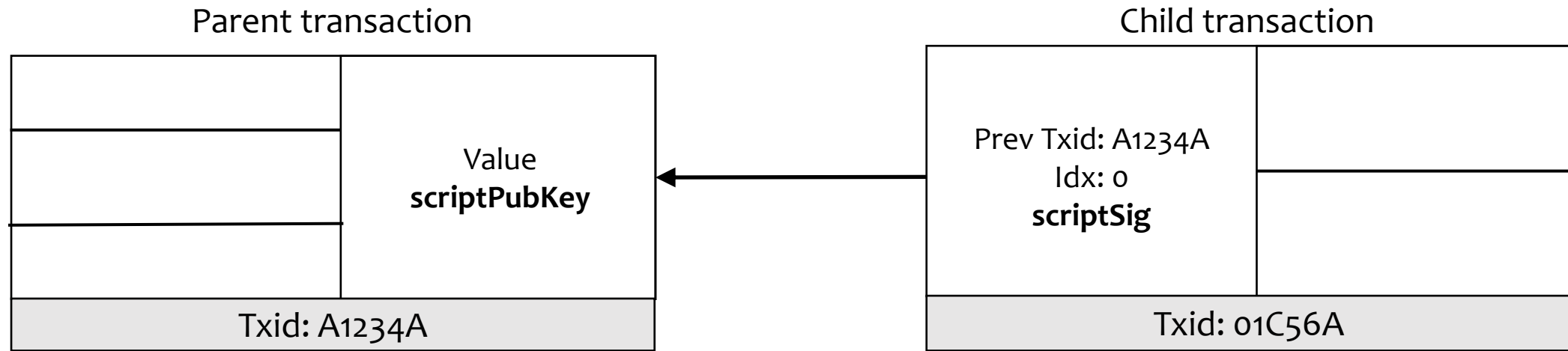
Rules for creating inputs and outputs

- Each output should mention a value and a 'scriptPubKey'
 - **scriptPubKey is the locking script that defines the spending condition for the output**
- Each input must reference exactly one output from a past txn (parent txn)
 - The input contains a 'scriptSig'
 - **scriptSig is the unlocking script for the referenced output in legacy transactions**
 - **scriptSig is empty in SegWit transactions (instead witness field contains the unlocking script)**
- The sum of input values should be greater than or equal to the sum of output values

With these rules, Bitcoin does not require user accounts

Spending an Output

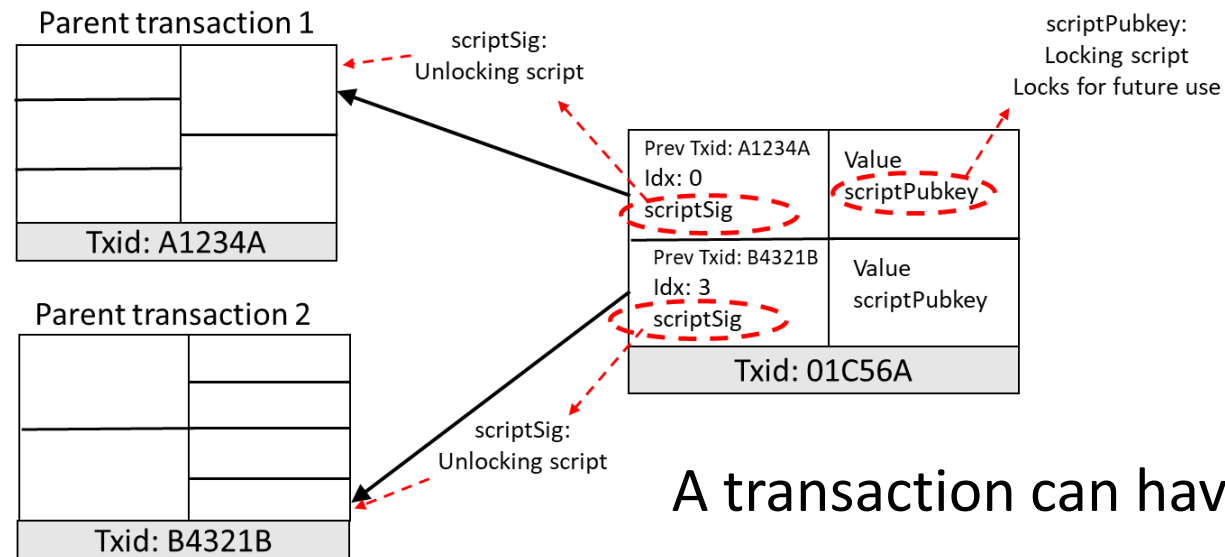
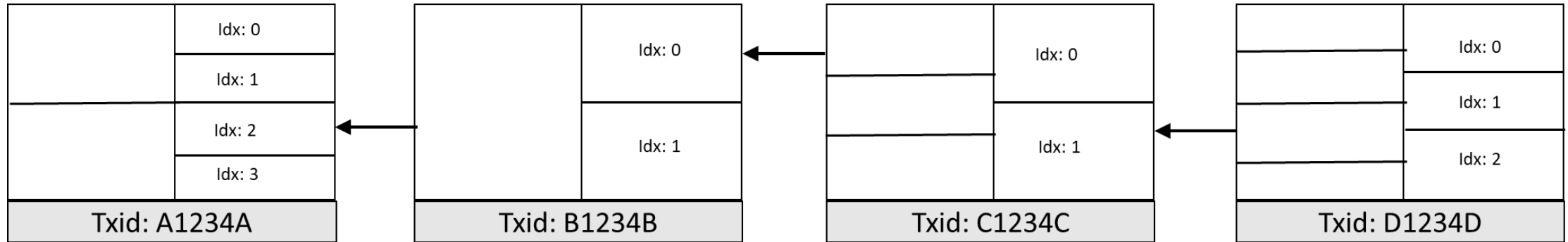
- Parent transaction is also referred to as the funding transaction
- Child transaction is also referred to as the spending transaction



- The value is locked at the **scriptPubKey** field at the output of the parent transaction
- The input of the child transaction identifies the output of the parent transaction that it wants to spend and provides the required **scriptSig** to unlock the output
- scriptSig + scriptPubKey is executed on the stack and if the result is TRUE, the output is unlocked and completely spent via the new outputs of the child transaction

Transaction Chaining

Bitcoin transactions also form a chain: each transaction spends output(s) of previous transaction(s) and creates new outputs for future use



A transaction can have multiple inputs

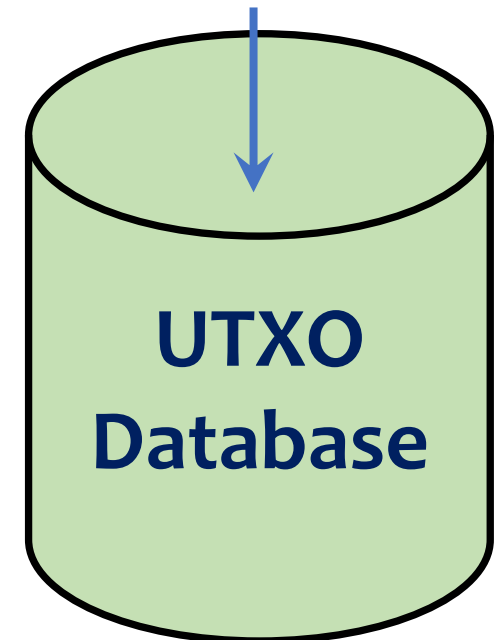
UTXO Database

- UTXOs are scattered in the blocks
 - Nodes maintain a UTXO database (called chainstate.db in Bitcoin core)
 - The purpose of this database is to speed up the transaction validation process

Simplified Transaction Validation & Database Update Algorithm

1. For each input in a new TXN (child txn):
 - a. If parent UTXO is not in the DB, return an error
 - b. If the provided signature on child txn does not match the owner (Pubkeyhash) of the parent UTXO, return an error
2. If the total value of all the referenced parent UTXOs is less than the total value of all the outputs of child txn, return an error
3. If successful, remove all parent UTXOs and add all child UTXOs to the database

Transaction outputs
that have not been
spent at a given
time

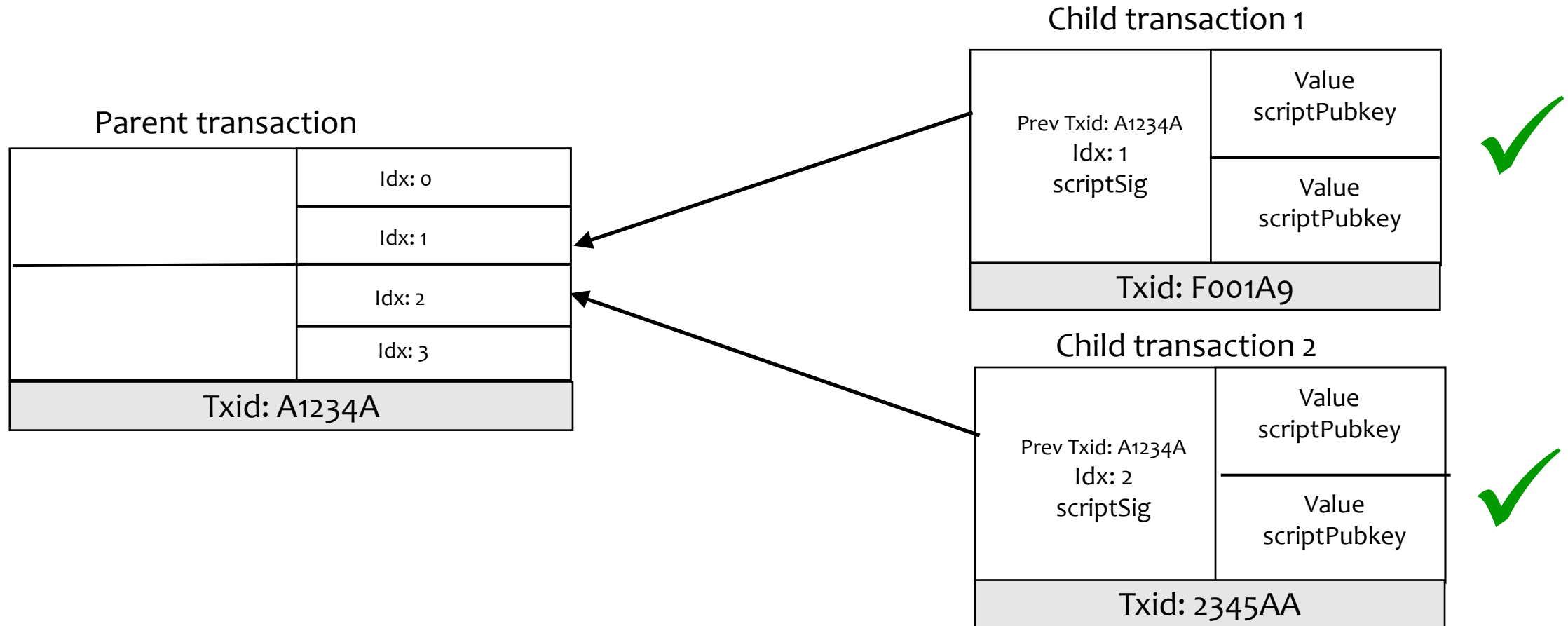


Lecture 14

Thursday, Mar 5, 2026

UTXO Model

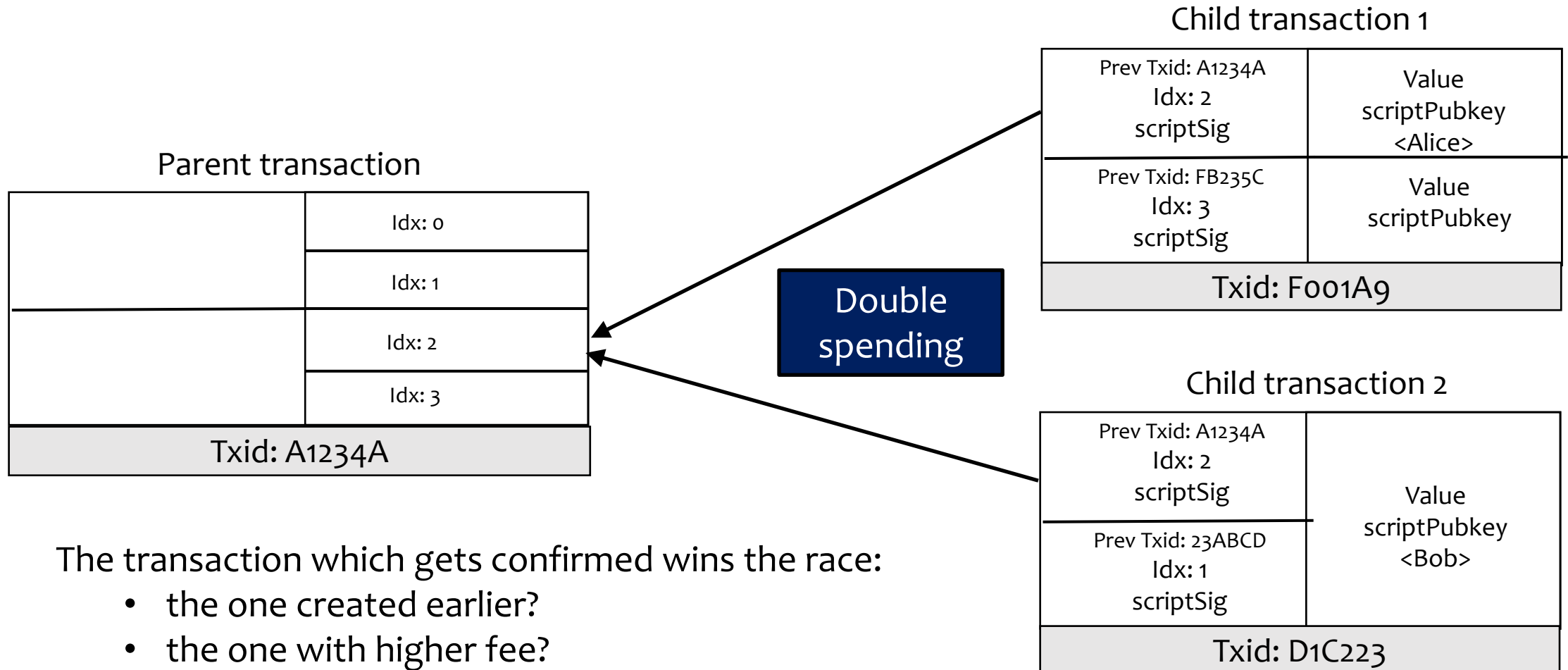
Which child txn is valid assuming signatures match and total value of outputs is less than the value of referenced UTXO?



Both are valid because they are pointing to two different outputs Idx 1 and Idx 2 respectively

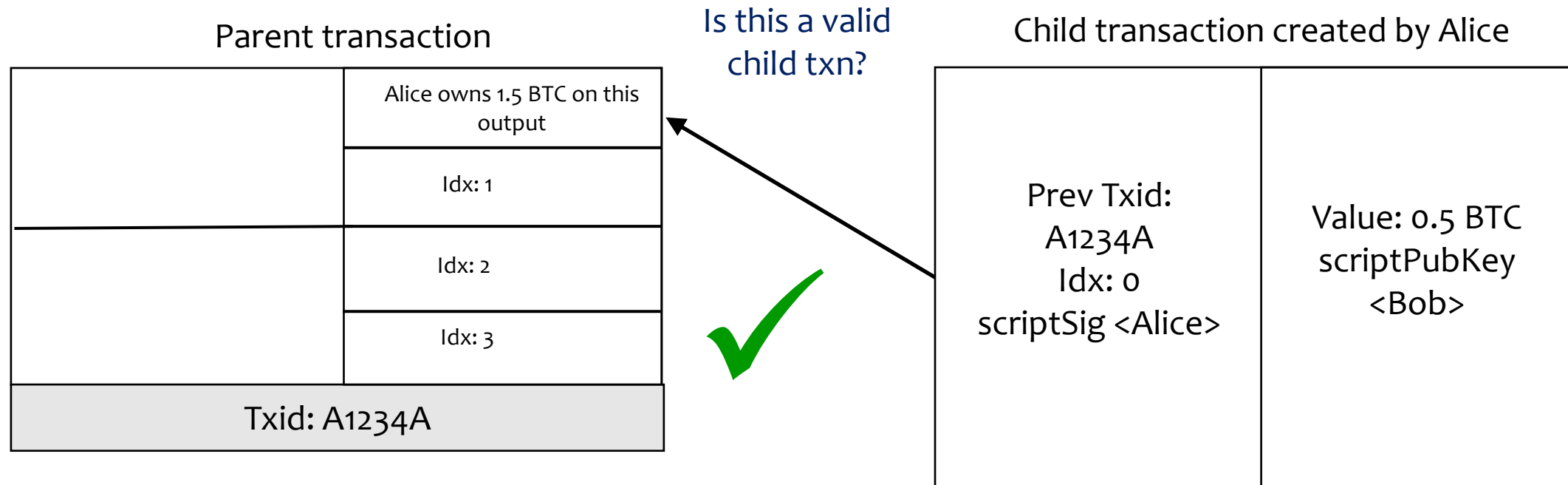
Race between Transactions

The two child transactions are in a so-called ‘race’



Making payments

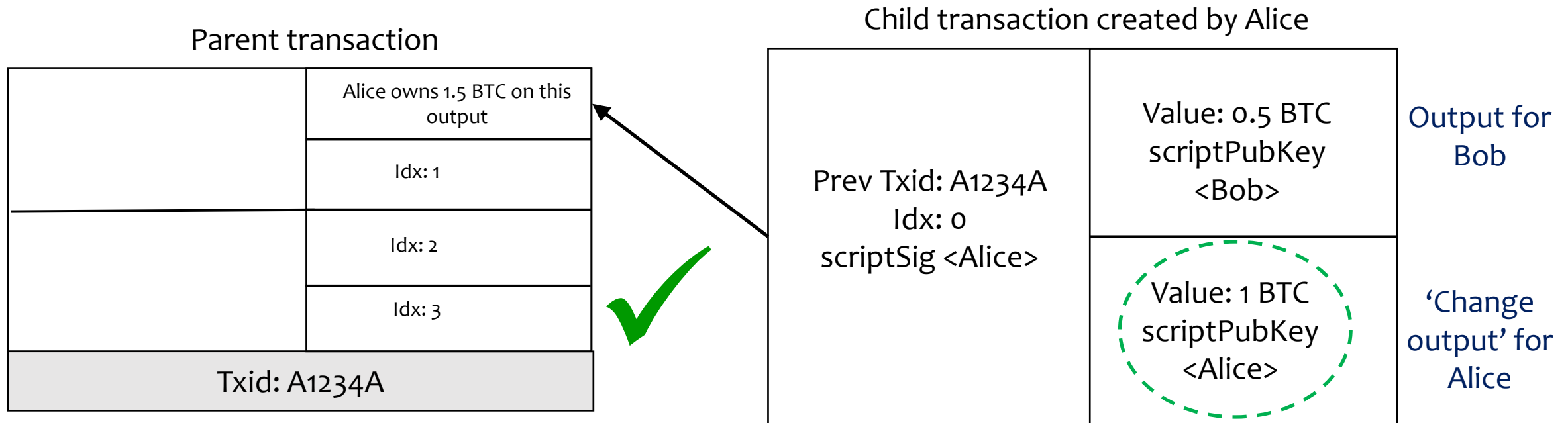
Alice wants to pay 0.5 BTC to Bob. Alice owns 1.5 BTC at index 0 of Txid: A1234A



Alice owns a valid UTXO, and sum of inputs > sum of outputs, so this is perfectly fine

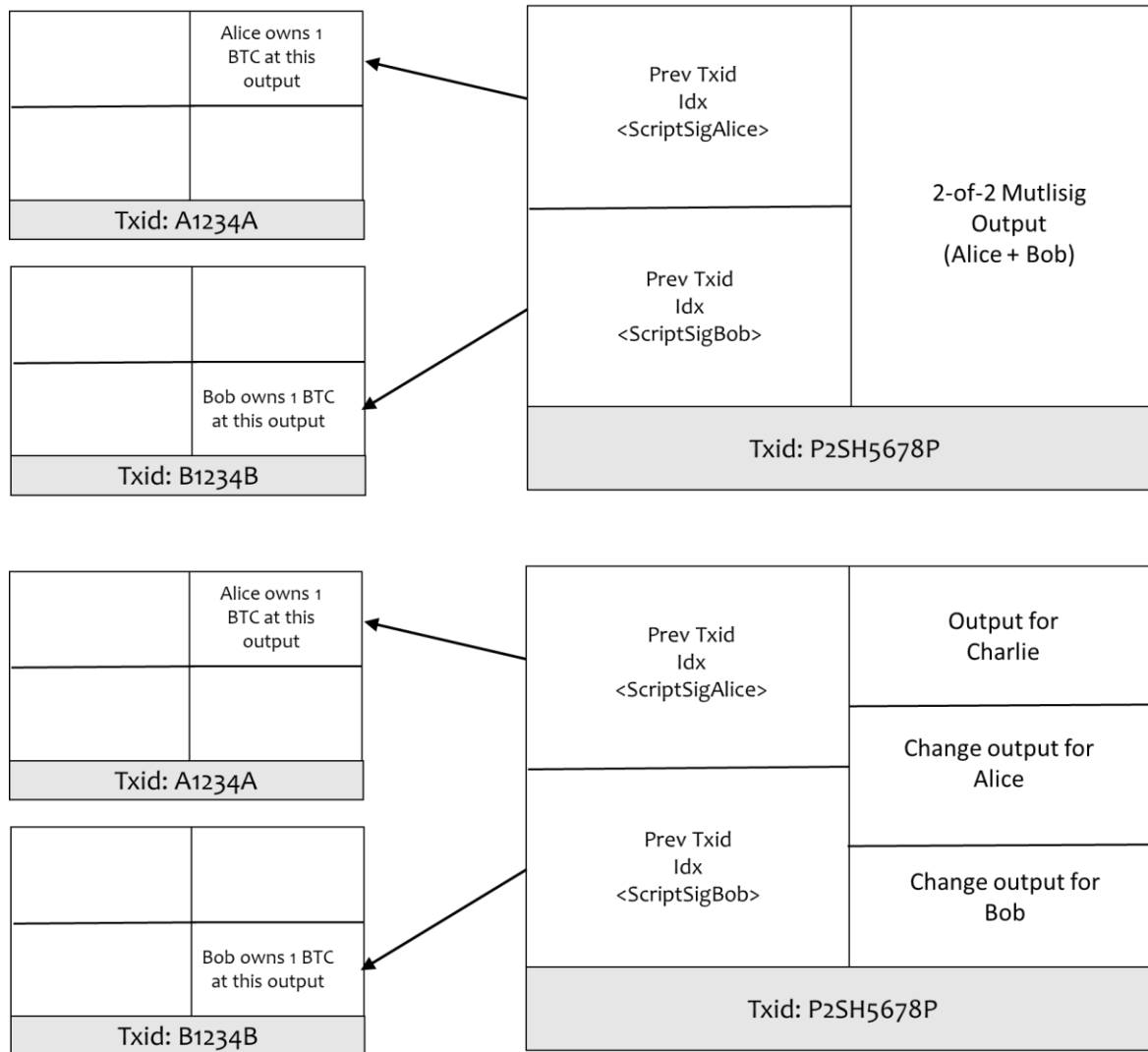
Any difference between the inputs and outputs is retained by the miner as txn fee so Alice will loose 1 BTC in this example

Payment with a 'Change Output'



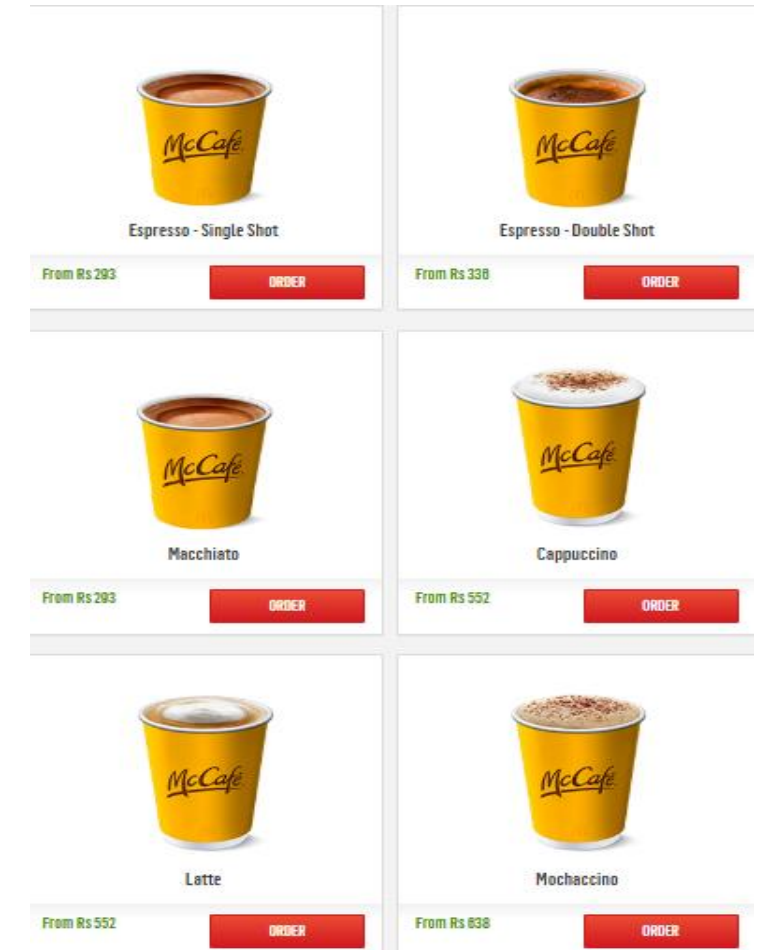
- Alice must pay herself the remaining BTC by creating a second output – this is equivalent to receiving a change for her future use
- This is a typical Bitcoin payment transaction with two outputs – one payment output to the intended recipient, and a change output returning the remaining funds to the sender

Collaborative Transaction



- Alice and Bob collaborate and agree on all the inputs and all the outputs (value, spending conditions)
- Either Alice or Bob creates an ‘*unsigned transaction*’
 - Unsigned transaction is shared with other party
- Alice signs the unsigned transaction (all the inputs and all the outputs) with her private key (produces <SigAlice> for this txn)
- Bob signs the unsigned transaction (all the inputs and all the outputs) with his private key (produces <SigBob> for this txn)
- To obtain a fully signed transaction, both parties exchange their signatures
 - The party holding all the signatures has the fully-signed transaction
 - The party with only some signatures holds a so-called partially signed bitcoin transaction (PSBT)
- Only the fully signed transaction can be broadcast to the Bitcoin network

BTC Price vs a Cup of Coffee



Bitcoin seems better suited for buying the entire coffee shop than a single cup!

Start Thinking in Satoshis!

- For micropayments, let us start thinking in Satoshis (sats)
- 1 BTC = 10^8 sats (100 million)
- Assume BTC price is 100,000 \$ (10^5 \$)
- 1 sat = $10^5 / 10^8 = 1/1000 = 0.001$ \$ (0.3 pkr)
- 1 cup of coffee is approximately 1 \$ or 1000 sats

Satoshi to USD / BTC Converter & Calculator

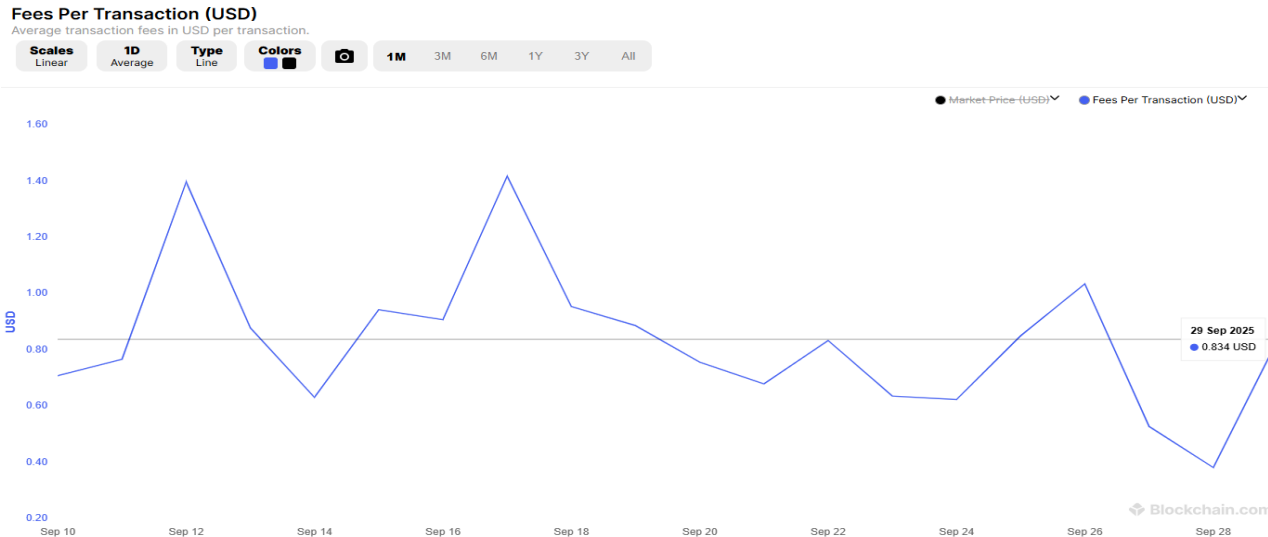
Currency	Amount
 Satoshi SATS	871
Currency	Amount
 US Dollar USD	1.00
Currency	Amount
 Bitcoin BTC	0.00000871

<https://bitbo.io/tools/satoshi-usd/>

Now bitcoin starts to feel like a usable currency

Transaction Fee

- 1 cup of coffee is approximately 1 \$ or 800 to 1000 sats
- Average transaction fee is around 0.5 to 1.5 \$ ~ 400 to 800 sats



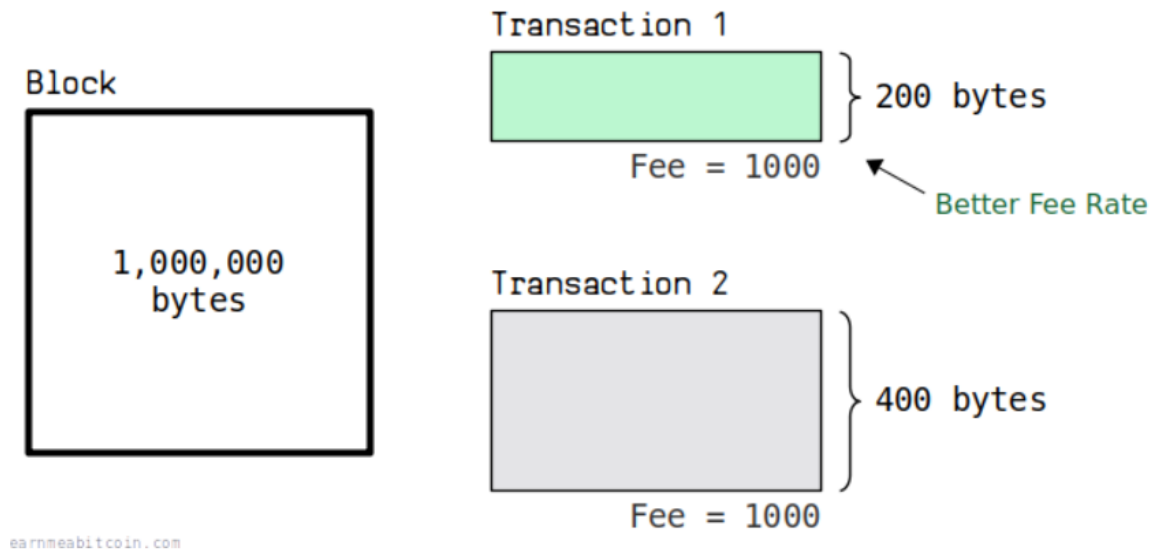
Bitcoin Transaction Cost Estimate

Transaction Type	Median sat/vB	Size [vB]	Cost [sats]
P2TR (Taproot) 2 Inputs + 2 Outputs	2.056	212	435
P2TR (Taproot) 2 Inputs + 3 Outputs	2.056	255	523
P2WPKH 1 Input + 2 Outputs	2.056	141	289
P2WPKH 2 Inputs + 2 Outputs	2.056	209	429
P2WPKH 2 Inputs + 3 Outputs	2.056	240	492
P2PKH 1 Input + 2 Outputs	2.056	226	465
P2PKH 2 Inputs + 2 Outputs	2.056	374	769
P2PKH 2 Inputs + 3 Outputs	2.056	408	839

It still seems difficult to use BTC for micropayments due to high transaction fee!

Transaction Fee Depends on Size

- In Bitcoin, transaction fee does not depend on the transaction value
- Transaction fee depends on the transaction size
 - Large sized transactions (more bytes) are costly
- A 1 \$ coffee payment transaction can end up paying the same fee as a 100,000 \$ BTC transfer if the transaction size in bytes is the same



Can we exploit this property to reduce the fees for micropayments?

Reducing the Fee for Micropayments

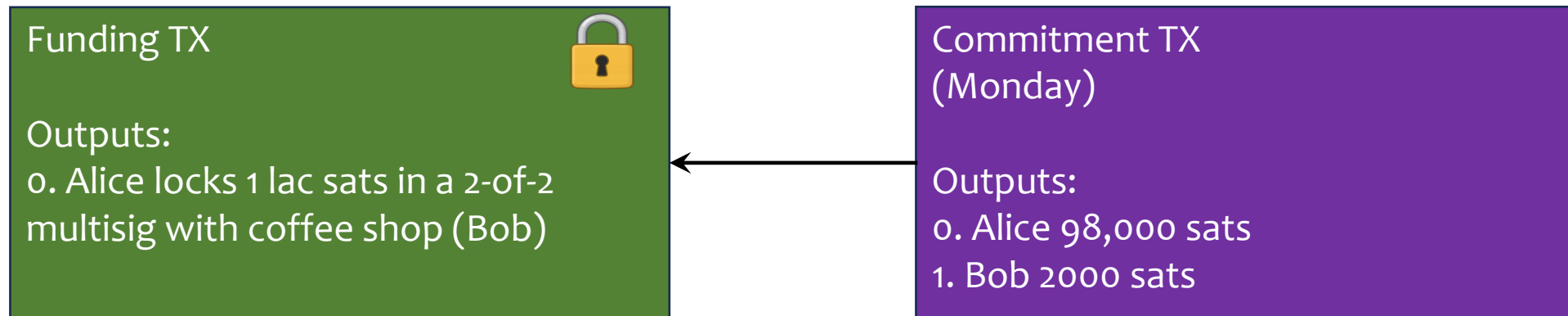
- Suppose we make a single high-value transaction with an output of 1 lac sats
 - This transaction is confirmed on blockchain
 - We pay the transaction fee (e.g., 500 sats)
 - We call it the ‘Funding transaction’
- We use the funding transaction to make several micropayments
 - After every micropayment, we only update the balances off-chain by creating ‘commitment transactions’
 - After several micropayments, we settle the final balances on-chain
 - We pay the transaction fee for the last commitment transaction that is confirmed on the blockchain (e.g., 500 sats fee)
- We pay the fee for the initial funding transaction and the last commitment transaction as they enter the bitcoin blockchain (only 1000 sats in transaction fees)

Commitment transactions exist off-chain, so the challenge is to make them as secure as the on-chain Bitcoin transaction

Funding & Commitment Transactions

Funding transaction:
2-of-2 multisig transaction on blockchain
used to lock funds in shared ownership

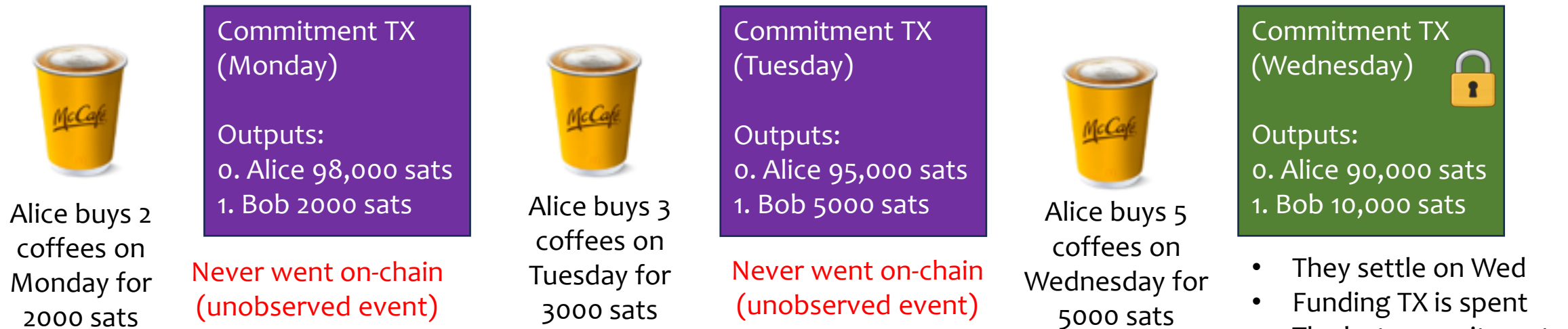
Commitment transaction:
A transaction or a number of transactions
that return the funds of a two-party
channel to their owner



- Before publishing the funding transaction, the funder first obtains a pre-signed spending transaction from the counterparty so the funds can always be recovered even if cooperation stops
- Once the funding transaction is safe to publish, the peers move forward by creating the commitment transactions

Unobserved Events

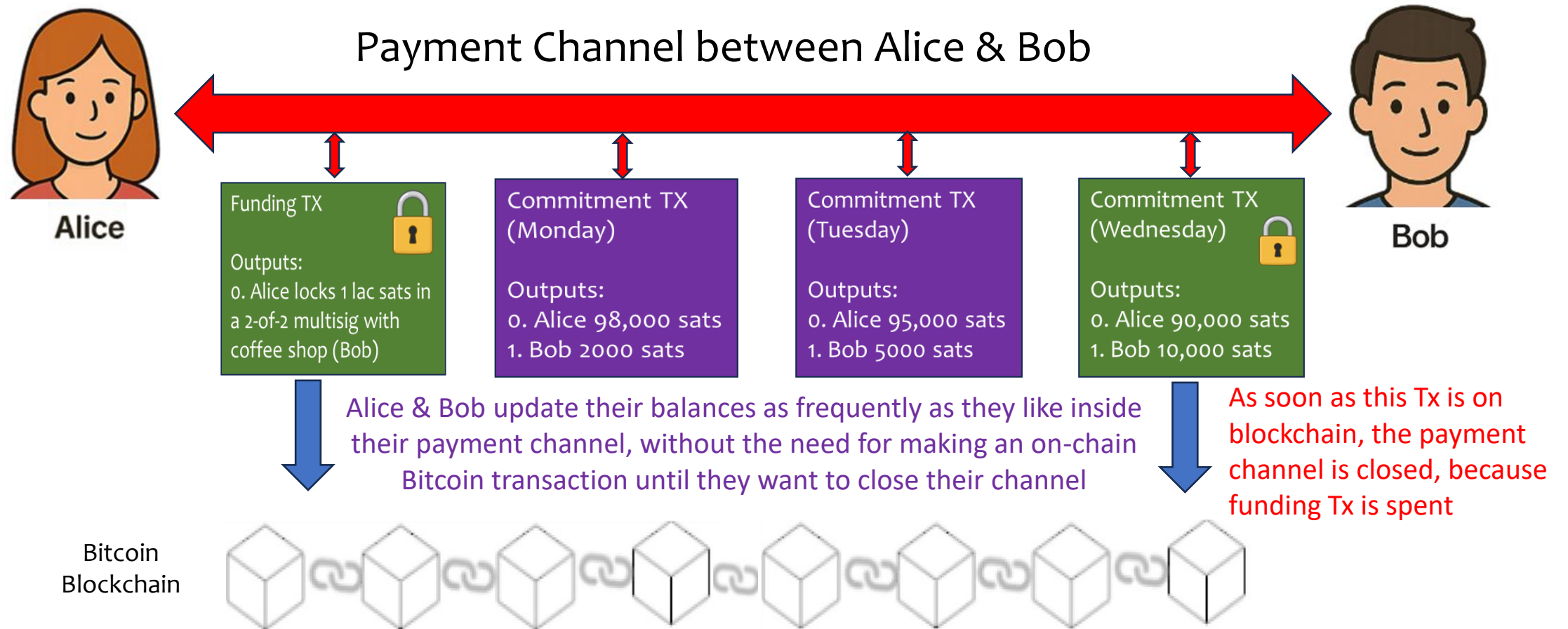
- After every coffee purchase, Alice & Bob create new commitment transactions and update their balances
- Either Alice or Bob can enforce the settlement by broadcasting and enforcing the latest agreed state without needing cooperation from the other party



All the intermediate transactions remain unobserved, which is also good for privacy

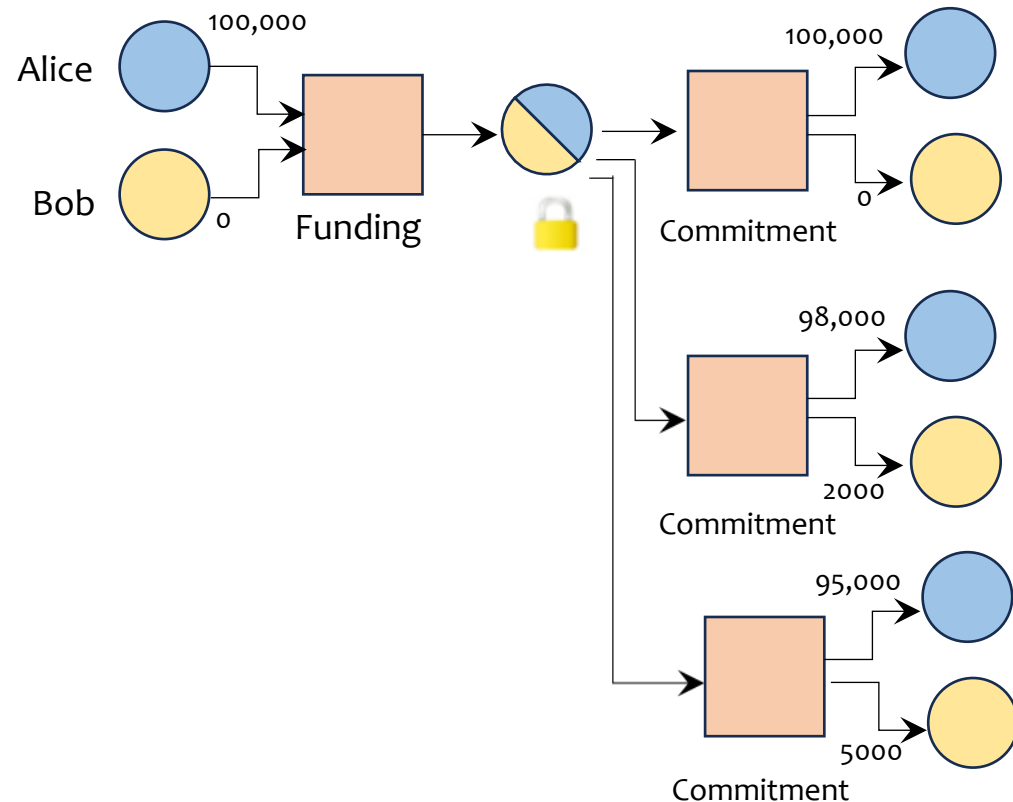
Payment Channel

Payment Channel is a setup that allows two parties to make payments without contacting the blockchain



Making this payment channel secure and trustless is a major challenge

Payment Channel



1. Alice & Bob decide to open a payment channel
2. Suppose Alice funds the channel with 100,000 sats
3. Alice & Bob create a 2-of-2 multisig funding tx
4. Bob signs a commitment transaction to pay back Alice
5. Once Alice has the commitment transaction, the funding transaction is broadcast and confirmed on-chain
6. Alice buys coffee and together with Bob decides to update the channel
7. Alice & Bob create a new commitment transaction with updated balances
8. Alice buys another coffee and together they decide to update the channel
9. They update their balances in the new commitment transaction
10. The process continues till they decide to close the channel